

Gamma detection for source localisation in targeted alpha therapy

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Abstract. Targeted alpha therapies can deliver a highly localised dose to tumours, but treatment planning and effectiveness depend on whether the radionuclide source remains near the intended site after administration. In the AlphaGlue delivery concept, alpha emitters are applied to a tumour region in a thin glue layer. We present a Geant4 simulation study showing that decay-chain gamma detection can distinguish retained, broadly diffused and locally displaced source states in an AlphaGlue-like geometry. 50 repeated simulations of 10,000 primary events were generated, and baseline-subtracted gamma-hit profiles were successfully used to reconstruct the broad diffusion fraction with $R^2 = 0.9987$ and a mean absolute error of about one percentage point. Localised-source residuals identified displaced activity with mean position errors of 16–30 mm. These results support gamma detection as a candidate method for emitter-position monitoring, treatment planning and delivery verification.

1 Introduction

In targeted alpha therapy, the delivered dose is strongly local. A treatment plan therefore depends on the initial activity distribution and its evolution after administration. The AlphaGlue concept proposes suspending alpha emitters in a thin glue layer applied to tissue [1]. The concept has previously been evaluated with Geant4 and Geant4-DNA simulations for dose and DNA-damage modelling [2–7]. Its delivered dose depends on whether source material and progeny remain near the intended deposition site, diffuse into surrounding tissue or are redistributed by biological transport [8,9]. This study investigates whether the accompanying gamma signal can be used as a theranostic observable to monitor emitter position, detect source movement and provide useful information as a function of detection time.

2 Simulation model

Photon transport was simulated with the Geant4 toolkit [2]. The model places a 40 mm × 2 mm × 40 mm loaded glue layer on the +y surface of a 20 cm × 20 cm × 150 cm water phantom. Six detector panels surround the phantom at 20 cm from the water surface. Each photon entry into a detector panel is counted as one hit.

Seven source configurations were studied to separate three clinically relevant possibilities: emitter retained in the loaded glue layer, emitter broadly mixed into the surrounding volume and emitter transported to a localised displaced region. The retained-source case is used as the baseline. Some alpha emitters in the decay chain can be taken up by the blood, with large variation reported between experiments [8,9]. This further emphasises the need for local dose verification. The simulations therefore include cases in which 10%, 50% or 90% of the radioisotope is distributed throughout the water volume, and cases in which 20%, 40% or 60%

of the source is placed in a 5 cm-side water cube centred at $(x, y, z) = (0, 0, 500 \text{ mm})$, spanning $475 \leq z \leq 525 \text{ mm}$. The geometry is described in Fig. 1, along with example detector displays showing the three clinical scenarios. Each configuration was simulated with 50 independent repetitions of 10,000 primary decays, giving 500,000 simulated decays per configuration and 3.5 million simulated decays across all source states. The repeated runs define the run-to-run uncertainty used in the residual and detectability analysis. The simulated decays are rescaled to a DaRT-like reference activity of $3 \mu\text{Ci}$ (111 kBq), matching the order of a single DaRT seed [10].

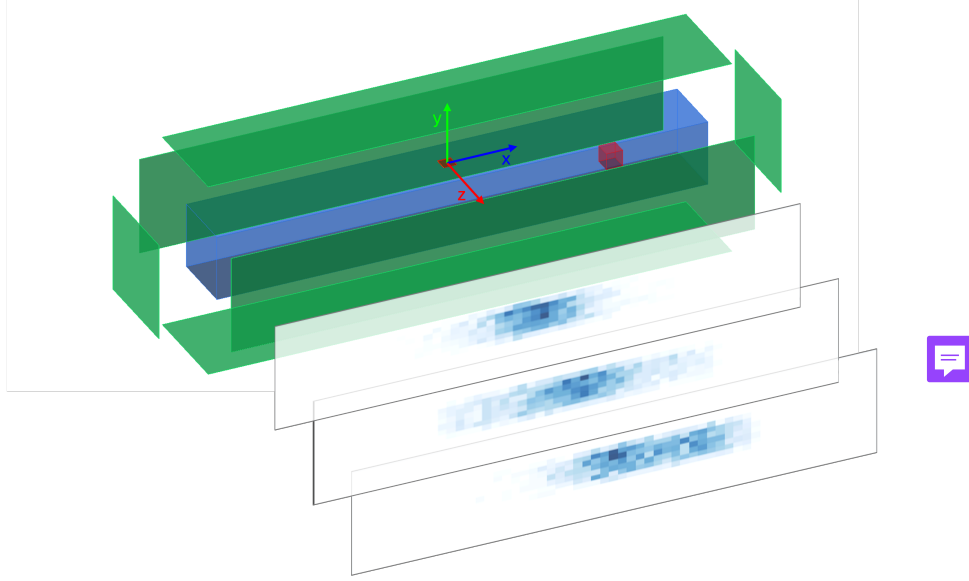


Figure 1: Simulation geometry showing the water phantom (blue), surrounding detector panels (green), a displaced source (red) and three offset example x -panel projections: baseline (top), broad redistribution (middle), and displaced localisation (bottom).

3 Baseline subtraction

The total detector-entry rate changed only weakly between retained and redistributed source states. Between the retained baseline and the 90% broad-diffusion case, the difference in hit count changed only by 3.8%. In contrast, baseline-subtracted profiles show a clear spatial signature. The retained source defines the reference profile $\bar{h}_p^B(z)$ for detector panel p . Candidate source states are represented by

$$\Delta h_p(z) = h_p(z) - \bar{h}_p^B(z), \quad (1)$$

with the repeated baseline simulations defining the bin-wise noise scale. As shown in Fig. 2, positive residuals mark detector regions where the candidate produces more gamma entries than the retained-source case, while negative residuals mark depleted regions. In the combined (x)-panel projection, broad diffusion widens the residual pattern, whereas localised displacement produces a shifted excess near the displaced source region. Changes in the balance between the glue-facing and opposite detector panels provide an additional feature for the quantitative analysis.

As an illustration of the general ROI strategy, Fig. 3 compares the whole x -panel profile with a region at $-750 \leq z \leq -550 \text{ mm}$, where little signal is expected from retained activity. An

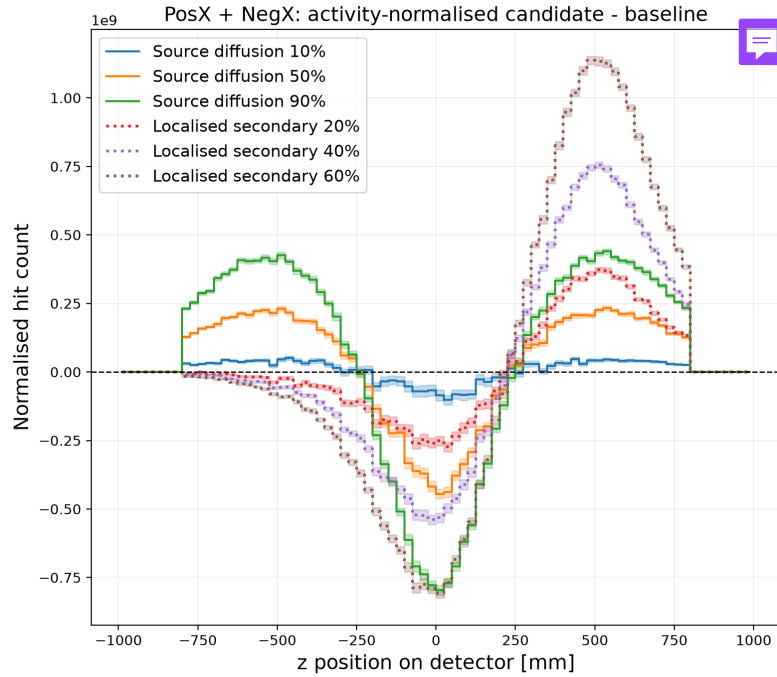


Figure 2: Activity-normalised, baseline-subtracted combined $+x$ and $-x$ detector profiles for broad diffusion and localised displacement. Shaded bands show variation across repetitions.

excess in this region indicates that the detected distribution has spread beyond the applicator footprint. This ROI illustrates the principle on which the quantitative estimators in Section 4 are based: source redistribution is inferred from spatial changes relative to the retained-source reference.

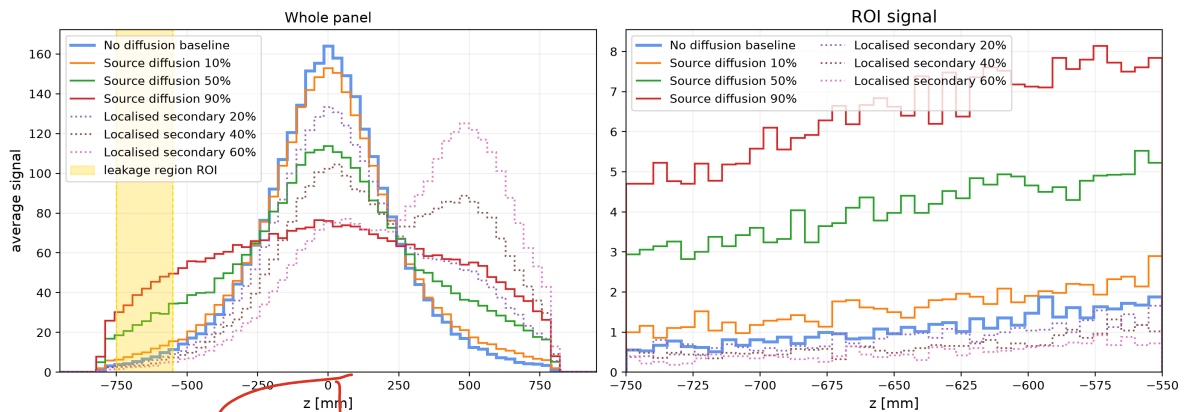


Figure 3: Whole x -panel profiles (left) and an illustrative low-signal ROI (right). The ROI highlights the broad-diffusion excess outside the retained-source region.

4 Signal analysis and source redistribution metrics

The signal analysis addresses four tasks: detecting and locating a compact secondary source, estimating the broadly diffused fraction when no compact source is present, estimating how

much signal remains consistent with the applicator region, and determining the observation time required to distinguish each candidate from the retained-source baseline. Both broad diffusion and compact displacement can reduce the central signal, so the compact-secondary test is applied first. A negative result enables the broad-diffusion and applicator-retention estimates; a positive result instead triggers source localisation.

4.1 Compact secondary-source detection and localisation

For each detector axis, let N_{+i} and N_{-i} denote the numbers of gamma entries recorded on the detector panels on the positive and negative sides of axis i , respectively, during the selected observation window. The panel asymmetry is

$$A_i = \frac{N_{+i} - N_{-i}}{N_{+i} + N_{-i}}, \quad i \in \{x, y, z\}. \quad (2)$$

Broad diffusion changes A_y , because activity moves away from the glue-facing panel, but remains approximately symmetric in x and z . A compact source displaced to positive z produces a large change in A_z . The primary compact-source test therefore compares (A_x, A_z) with the retained-source mean using the **squared Mahalanobis distance**, with the covariance estimated from the baseline repetitions. A balanced logistic-regression classifier using standardised asymmetry, positive residual, centroid and ROI-rate features was used as a cross-check with leave-one-repetition-out validation.

As a complementary test, the spatial distribution of hits within the detector panels is examined in a region surrounding the expected displaced-source position. For this geometry the one-dimensional ROI is $250 \leq z \leq 750$ mm in the **combined** $+x/-x$ and $+y/-y$ views; the corresponding two-dimensional ROI on an x panel is $-100 \leq y \leq 100$ mm and $250 \leq z \leq 750$ mm. The detection score is the sum of positive candidate-minus-baseline residuals in the ROI, normalised by the spread of the same score across baseline repetitions. After a positive detection, the source position is estimated from the positive-residual centroid in the combined x -panel projection,

$$\hat{z}_{\text{sec}} = \frac{\sum_{z \in \text{ROI}} z \max[\Delta h^{X\text{sum}}(z), 0]}{\sum_{z \in \text{ROI}} \max[\Delta h^{X\text{sum}}(z), 0]}. \quad (3)$$

The reconstructed positions and their repetition-level residuals are reported in Fig. 6.

4.2 Broad-diffusion fraction

Broad diffusion is estimated only after the compact-secondary test is negative. Inputs comprise shape-normalised z -profile bins from the combined x , combined y and all-panel views; the standard deviation and central-80% interquantile width ($z_{90} - z_{10}$); the central $+y$ fraction; x - and y -panel asymmetries; and positive-residual integral and centroid summaries. These features are standardised and supplied to ridge regression. The regularisation strength is selected with **RidgeCV** over $\alpha = 10^{-4}$ – 10^4 . The model is trained only on the retained and 10%, 50% and 90% broad-diffusion states; localised-source cases are excluded. Performance is evaluated with leave-one-repetition-out validation. The resulting predictions are shown in Fig. 5.

4.3 Applicator-retained signal

The applicator-facing observable is the $+y$ -panel count in the central ROI, $|z| \leq 150$ mm, normalised per primary event. Dividing this rate by the retained-source rate gives a baseline-relative proxy for the fraction of signal remaining at the applicator. The all-panel central ROI is retained as a cross-check. As shown in Fig. 4, both broad diffusion and compact displacement

reduce this signal, so an applicator-retained fraction is quoted only after the compact-secondary test has been evaluated.

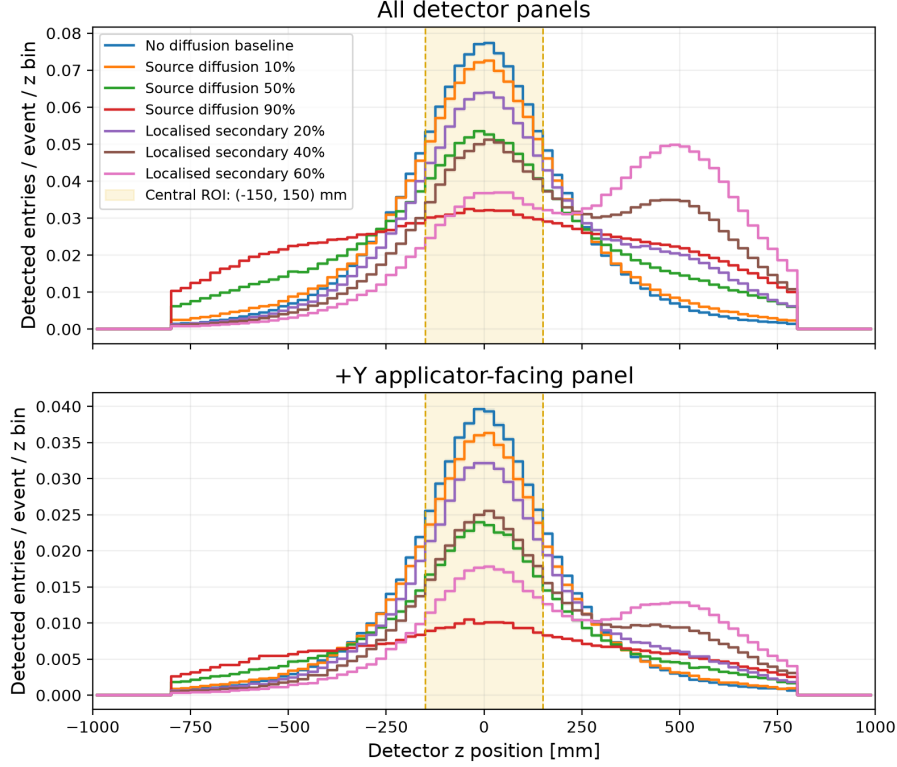


Figure 4: Central $|z| \leq 150$ mm ROI for all detector panels (top) and the applicator-facing $+y$ panel (bottom). The $+y$ ROI rate provides the baseline-relative applicator-retained signal.

4.4 Time-window detectability

The same baseline-subtracted profiles and task-specific ROIs are accumulated in increasing observation windows. At each window, the candidate signal is compared with the distribution obtained from the baseline repetitions. A source state is considered detectable when the candidate–baseline separation exceeds three standard deviations of the run-to-run baseline variation. The reported detection time is the first cumulative window that crosses this threshold. This procedure tests when the spatial redistribution used by the diffusion and localisation estimators becomes statistically resolvable; the resulting threshold times and an example cumulative profile are presented in Section 5.2.

5 Results

5.1 Signal inference

Using profile-shape features from repeated simulations, the broad diffusion fraction was recovered with $R^2 = 0.9987$ and a mean absolute error of about one percentage point (Fig. 5). The baseline-normalised central $+y$ -panel ROI signal decreased monotonically with broad diffusion, to 0.925 ± 0.017 , 0.618 ± 0.013 and 0.289 ± 0.009 for the 10%, 50% and 90% broad-diffusion cases, respectively (Fig. 4). The ratio measures the central-ROI detector signal relative to the retained-source baseline; it is not a direct measurement of the fraction of activity remaining in

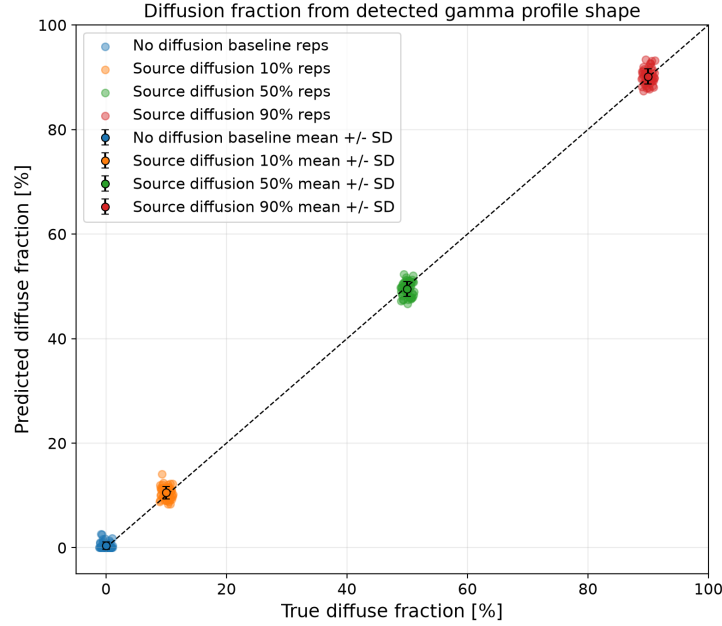


Figure 5: Diffusion fraction inferred from 50 independent repetitions. Small points show individual repetitions and large markers show the mean and standard deviation.

the applicator. The compact-source strategy identified the localised displaced source in all 20%, 40% and 60% localised-displacement cases. The reconstructed displaced-source positions were approximately $z = 516$ mm, 528 mm and 530 mm, respectively, compared with the simulated source centre at $z = 500$ mm. Across these localised fractions, mean position errors were 16–30 mm, with the repetition-level residual distributions shown in Fig. 6. The repetition-to-repetition spread decreased from 6.6 mm to 2.1 mm as the displaced-source fraction increased, while the mean estimate remained shifted towards positive z . This indicates increasing precision but a systematic bias in the positive-residual centroid estimator; calibration using simulations at multiple source positions will be required in future work.

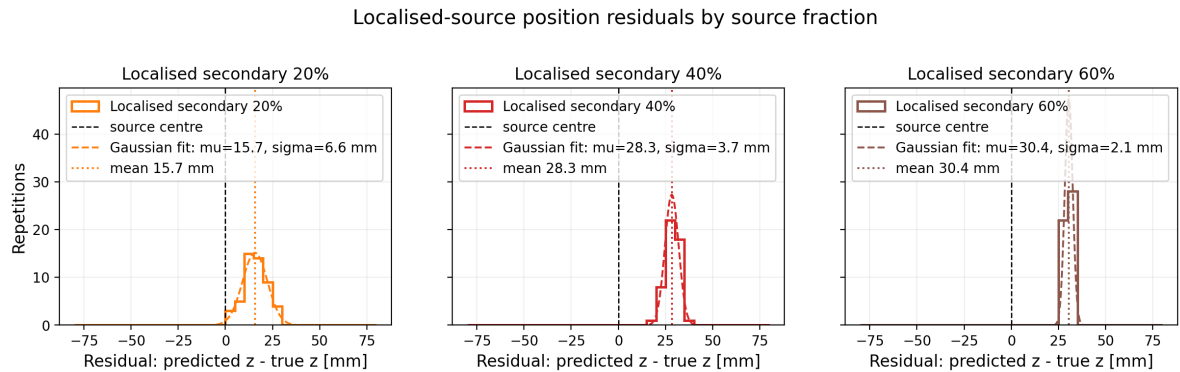


Figure 6: Localisation residuals across 50 repetitions for each displaced-source fraction. The dashed black line marks the true source centre; coloured lines indicate the mean and Gaussian fit.

5.2 Time dependence

Cumulative time-window profiles show that the baseline-subtracted signal is already visible on hour-scale windows and becomes stronger over longer observation windows. Across the 50 simulation repetitions, detectability was defined as candidate-baseline separation greater than 3σ of the run-to-run baseline variation. The 50–90% broad-diffusion and 20–60% localised-displacement cases crossed threshold in about 1–2 hours, while the weaker 10% broad-diffusion case required about 6 hours. A representative time-window profile is shown in Fig. 7. A $t < 10^6$ s analysis cut suppresses a late decay-chain tail while retaining the early signal relevant to the simulated source-localisation window. The first threshold-crossing times are summarised in Table 1.

Table 1: First 3σ detection times from the time-window analysis of 50 independent repetitions.

Case group	First 3σ detection
10% broad diffusion	about 6 h
50–90% broad diffusion	about 1–2 h
20–60% localised displacement	about 1–2 h

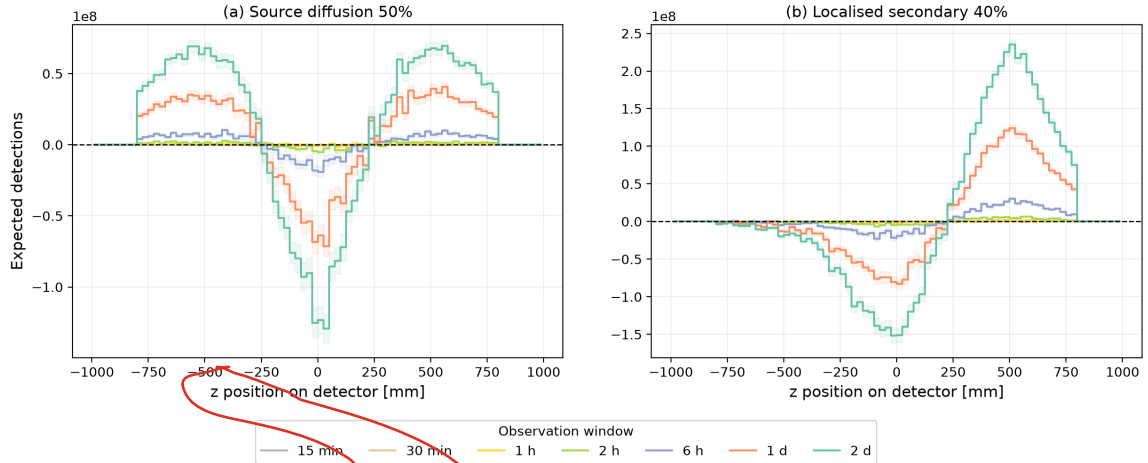


Figure 7: Cumulative combined $+x/-x$ residual profiles for 50% broad diffusion (left) and 40% localised displacement (right). Broad diffusion produces a symmetric redistribution away from the centre, whereas localised displacement produces an asymmetric excess near the displaced-source region. These signatures first exceed the 3σ detection threshold after approximately 2 hours and 1 hour, respectively.

6 Conclusion

Decay-chain gamma detection provides source-location information and broad redistribution estimation. Baseline-subtracted profiles distinguish retained, broadly diffused and locally displaced activity, while residual features support quantitative estimates of diffusion fraction and displaced source position. Broad diffusion and compact source displacement were treated as independent source states in the present analysis. A combined case, in which diffuse redistribution and a compact displaced source occur together, should be included in the next simulation and

inference studies. Further work will also include detector response, background estimation and patient-scale geometries so that activity-normalised profiles can be interpreted as experimentally measurable count rates.

References

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